

Cascaded Channel Estimation for Distributed IRS Aided mmWave Massive MIMO Systems

Group Meeting, SPC Lab, ECE, IISc

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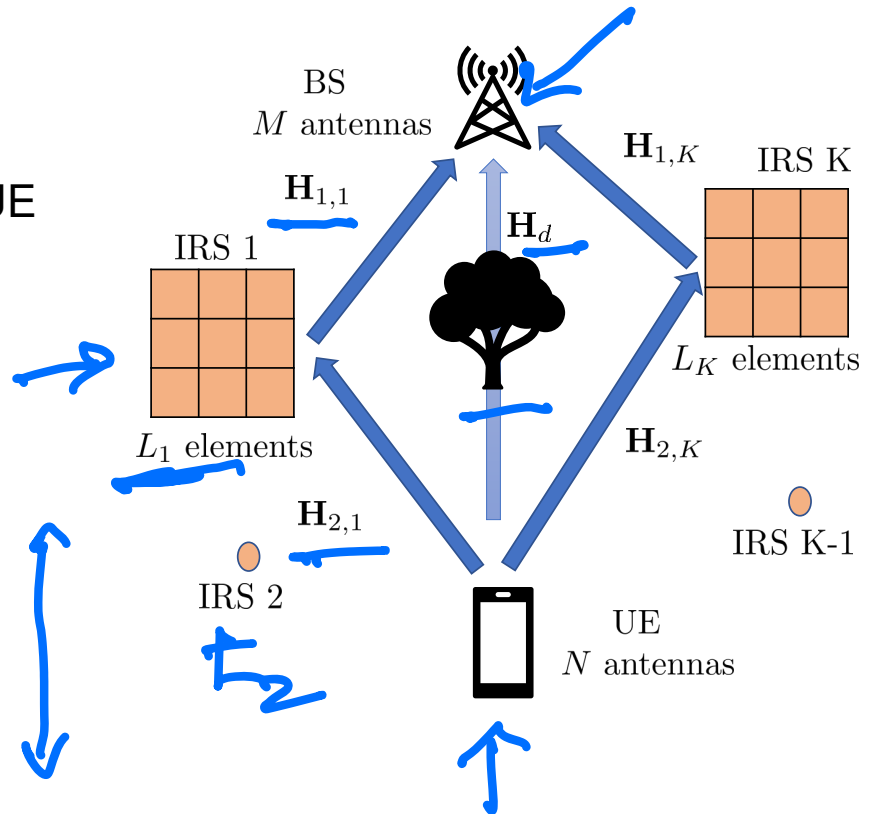
Today...

Explore channel estimation for distributed IRS aided mmWave massive MIMO systems

- Propose a novel cascaded channel estimation technique (in uplink):
 - Low pilot overhead
 - Account for channel structures
 - Leverage array processing tools - propose a *joint ESPRIT-MUSIC algorithm*
- Analyze and compare the performance with existing approaches

Distributed IRS aided mmWave mMIMO Model

- 1 M -antenna BS; N -antenna UE
- 2 K IRSs (configured as ULA) with L_k elements each
- 3 Saleh-Valenzuela model for channel modeling
- 4 Channel components:
 - Direct channel - $\mathbf{H}_d \in \mathbb{C}^{M \times N}$
 - UE to k th IRS - $\mathbf{H}_{2,k} \in \mathbb{C}^{L_k \times N}$
 - k th IRS to BS - $\mathbf{H}_{1,k} \in \mathbb{C}^{M \times L_k}$
 - Reflection coefficient at k th IRS - $\boldsymbol{\Theta}_k \in \mathbb{C}^{L_k \times L_k}$



Channel Modeling

Specifically, the links via the k th IRS are modelled as

$$\mathbf{H}_{2,k} = \sum_{i=1}^{N_k} \alpha_{i,k} \mathbf{a}_{L_k}(\omega_{i,k}) \mathbf{a}_N^T(\phi_{i,k}), \quad \forall k \in [K],$$

and

$$\mathbf{H}_{1,k} = \beta_k \mathbf{a}_M(\psi_k) \mathbf{a}_{L_k}^T(\nu_k), \quad \forall k \in [K],$$

where N_k - # spatial paths b/w UE and k th IRS and

$$\mathbf{a}_X(\theta) = [1, e^{-j\pi \sin(\theta)}, \dots, e^{-j(X-1)\pi \sin(\theta)}]^T,$$

Effective uplink channel:

$$\mathbb{C}^{M \times N} \ni \mathbf{H} = \mathbf{H}_d + \underbrace{\sum_{k=1}^K \mathbf{H}_{1,k} \mathbf{\Theta}_k \mathbf{H}_{2,k}}_{\triangleq \mathbf{G}}.$$

$\mathbf{\Theta}_k = \mathbf{I}$

Cascaded channel estimation design

Pre-processing steps

- Uplink received signal: $\mathbf{y}' = \mathbf{H}\mathbf{x} + \mathbf{n}'$
- Residual signal after direct component subtraction: $\mathbf{y} = \mathbf{G}\mathbf{x} + \mathbf{n}$
- Perform T_1 such pilot transmissions:¹ $\mathbf{Y} = \mathbf{G}\mathbf{X} + \mathbf{N}$ ($\mathbf{X} \in \mathbb{C}^{N \times T_1}$ – unitary)
- Matched filtering at the receiver: $\mathbf{Z} \triangleq \mathbf{Y}\mathbf{X}^H = \mathbf{G} + \mathbf{N}$

$$\mathbf{y} = \mathbf{y}' - \mathbf{H}_d \mathbf{x}$$

$$T_1 = N$$

Structure in the channel: Let $\Theta_k = \mathbf{I}_{L_k}$ for $k \in [K]$

$$\begin{aligned}
 \mathbf{G} &= \sum_{k=1}^K \mathbf{H}_{1,k} \mathbf{H}_{2,k} \\
 &= \sum_{k=1}^K \sum_{i=1}^{N_k} \alpha_{i,k} \beta_k \mathbf{a}_M(\psi_k) \underbrace{\mathbf{a}_{L_k}^T(\nu_k) \mathbf{a}_{L_k}(\omega_{i,k})}_{\triangleq \epsilon_{i,k}} \mathbf{a}_N^T(\phi_{i,k}) \\
 &= \sum_{k=1}^K \sum_{i=1}^{N_k} \underbrace{\alpha_{i,k} \beta_k \epsilon_{i,k}}_{\triangleq \gamma_{i,k}} \mathbf{a}_M(\psi_k) \mathbf{a}_N^T(\phi_{i,k})
 \end{aligned}$$

¹ To ensure orthogonality, we need $T_1 \geq N$.

Cascaded channel estimation design contd...

Let

$$\tilde{\mathbf{A}}_M \triangleq [\mathbf{a}_M(\psi_1) \odot \mathbf{1}_{N_1}^T, \dots, \mathbf{a}_M(\psi_K) \odot \mathbf{1}_{N_K}^T] \in \mathbb{C}^{M \times \sum_{i=1}^K N_i}$$

$$\mathbf{A}_N \triangleq [\mathbf{a}_N(\phi_{1,1}), \dots, \mathbf{a}_N(\phi_{N_1,1}), \mathbf{a}_N(\phi_{1,2}), \dots, \mathbf{a}_N(\phi_{N_2,2}),$$

$$\dots, \mathbf{a}_N(\phi_{1,K}), \dots, \mathbf{a}_N(\phi_{N_K,K})] \in \mathbb{C}^{N \times \sum_{i=1}^K N_i}.$$

Then, $\mathbf{G}$ is written as

$$\mathbf{G} = \tilde{\mathbf{A}}_M \mathbf{\Gamma} \mathbf{A}_N^T,$$

where $\mathbf{\Gamma}$: diagonal matrix of size $\sum_{i=1}^K N_i \times \sum_{i=1}^K N_i$.

$$\mathbf{Z} = \tilde{\mathbf{A}}_M \mathbf{\Gamma} \mathbf{A}_N^T + \mathbf{N}.$$

Vectorizing to get a **snapshot** of observations

$$\mathbf{z} = \mathbf{A}_{\text{eff}} \boldsymbol{\gamma} + \mathbf{n},$$

where $\mathbf{z} = \text{vec}(\mathbf{Z})$, $\boldsymbol{\gamma} = \text{diag}(\mathbf{\Gamma})$, $\mathbf{A}_{\text{eff}} = \mathbf{A}_N \odot \tilde{\mathbf{A}}_M \in \mathbb{C}^{MN \times \sum_{i=1}^K N_i}$ and $\mathbf{n} = \text{vec}(\mathbf{N})$.

Cascaded channel estimation design contd...

$$\mathbf{z} = \mathbf{A}_{\text{eff}} \gamma + \mathbf{n},$$

Handwritten annotations: $MN \times 1$ (pointing to $\mathbf{z}$), $MN \times EN_i$ (pointing to $\mathbf{A}_{\text{eff}}$), $EN_i \times 1$ (pointing to γ), and $MN \times 1$ (pointing to $\mathbf{n}$).

-An array signal model for a (virtual) array of MN sensors

- Consider as a direction finding problem:

- Estimate DoAs at the BS first using **ESPRIT**²
- Estimate DoDs at the UEs then jointly using **MUSIC**
- Finally, estimate $\hat{\mathbf{A}}_{\text{eff}}$

- Least squares to estimate the path-gain coefficients:

$$\hat{\gamma} = \hat{\mathbf{A}}_{\text{eff}}^{\dagger} \mathbf{z}_1 = (\hat{\mathbf{A}}_{\text{eff}}^H \hat{\mathbf{A}}_{\text{eff}})^{-1} \hat{\mathbf{A}}_{\text{eff}}^H \mathbf{z}_1.$$

- Cascaded channel estimate $\hat{\mathbf{G}}$: estimated using $[\hat{\mathbf{A}}_{\text{eff}}, \hat{\gamma}]$.

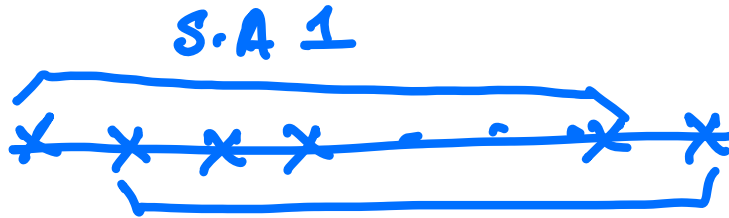
² Collect T_2 snapshots and estimate the data covariance matrix, $\hat{\mathbf{R}} \triangleq \frac{1}{T_2} \sum_t \mathbf{z}_t \mathbf{z}_t^H$. For independent snapshots, enable the IRSs with $T_2 - 1$ random phase configurations and 1st snapshot obtained with $\Theta_k = \mathbf{I}_{L_k} \forall k \in [K]$.

$$\mathbf{z}_1 \mathbf{z}_2 \dots \mathbf{z}_{T_2}$$

$$\mathbf{R} = \mathbf{a}_1 \mathbf{a}_1^H + \mathbf{b}_1 \mathbf{b}_1^H$$

Handwritten note: $\rightarrow 2$

Short background on ESPRIT



$$R(E_s) = R(A)$$

$$W_1 = \begin{bmatrix} I_{N-1} & 0 \\ 0 & 1 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} 0 & 1 \\ 1 & I_{N-1} \end{bmatrix}$$

SA 2

$$\underline{y}_1 = W_1 \underline{y} = \underline{W}_1 A \underline{x} + \underline{W}_1 \underline{\eta}$$

$$\underline{y}_2 = W_2 \underline{y} = \underline{W}_2 A \underline{x} + \underline{W}_2 \underline{\eta}$$

$$A_2 = A_1 \Phi$$

$$\Phi = \begin{bmatrix} e^{-j\pi \sin \theta_1} & 0 \\ 0 & e^{j\pi \sin \theta_2} \\ 0 & 0 \\ 0 & e^{-j\pi \sin \theta_p} \end{bmatrix}$$

$$R(A) = R(E_s)$$

$$A_1 = E_{s_1} F$$

$$A_2 = E_{s_2} F$$

$$E_{s_2} F = E_{s_1} F \Phi$$

$$\Rightarrow E_{s_2} = E_{s_1} F \Phi F^{-1}$$

$$\text{Sig. val } (E_{s_1}, E_{s_2}) = \text{E-val } (F \Phi F^{-1}) = \text{Eig-val } (\Phi) = \text{diag}(\Phi)$$

Short background on MUSIC

$$\text{EVD of } R_y = \underline{E} \underline{\Lambda} \underline{E}^H$$

$$\underline{E} = [\underline{E}_S : \underline{E}_N]$$

$$\underline{R}_y = [\underline{E}_S : \underline{E}_N] \begin{bmatrix} \underline{\Lambda}_S + \sigma^2 \underline{I}_p & 0 \\ 0 & \sigma^2 \underline{I}_{N-p} \end{bmatrix}; \quad \underline{\Lambda}_S = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_p \end{bmatrix}$$

$$\underline{E}_S^H \underline{E}_N = \underline{0} \Rightarrow \underline{e}_k \perp \underline{e}_l; l=1, \dots, p$$

$$\Rightarrow \underline{e}_k \perp \text{span}\{\underline{e}_l\}_{l=1}^p, \quad k=p+1, \dots, N \Rightarrow \text{span}\{\underline{e}_k\}_{k=p+1}^N \perp \text{R}(\underline{E}_S)$$

$$\boxed{\text{R}(\underline{E}_N) \perp \text{R}(\underline{A})}$$

$$\underline{R}_y = \underline{A} \underline{R}_z \underline{A}^H + \underline{R}_n = \underline{A} \underline{\Lambda}_z \underline{A}^H + \sigma^2 \underline{I}$$

$$\underline{\Lambda} = \begin{bmatrix} \lambda_1 + \sigma^2 & & \\ & \lambda_2 + \sigma^2 & \\ & & \ddots \\ & & & \lambda_p + \sigma^2 \\ & & & & \sigma^2 \\ & & & & & \ddots \\ & & & & & & \sigma^2 \end{bmatrix}$$

Estimation of DoAs at the BS using **ESPRIT**

Input: $\hat{\mathbf{R}}, M, N, K$

- 1 Perform Eigenvalue decomposition of $\hat{\mathbf{R}} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^H$ and obtain $(\mathbf{\Lambda}, \mathbf{U})$
- 2 Select the K dominant eigen vectors of $\mathbf{U}$ to form a basis for signal sub-space, $\mathbf{U}_S$
- 3 Define selection matrices $\mathbf{W}_1 \triangleq [\mathbf{I}_{MN-1} \mid \mathbf{0}_{MN-1}]$ and $\mathbf{W}_2 \triangleq [\mathbf{0}_{MN-1} \mid \mathbf{I}_{MN-1}]$
- 4 Compute $\mathbf{U}_{S,1} = \mathbf{W}_1 \mathbf{U}_S$ and $\mathbf{U}_{S,2} = \mathbf{W}_2 \mathbf{U}_S$
- 5 Obtain $\mathbf{\Omega} = \mathbf{U}_{S,1}^\dagger \mathbf{U}_{S,2}$
- 6 Compute the eigen values of $\mathbf{\Omega}$ as $\xi_1, \xi_2, \xi_3, \dots, \xi_K$
- 7 Compute $\hat{\psi}_k = \sin^{-1} \left(-\frac{\arg(\xi_k)}{\pi} \right)$, where $\arg(x) = \tan^{-1} \left(\frac{\Im(x)}{\Re(x)} \right)$, $k = 1, \dots, K$

Output: $\{\hat{\psi}_k\}_{k=1}^K$

Algorithm 1: DoA estimation at the BS using ESPRIT

Joint estimation of DoDs at the UEs using **MUSIC**

Input: $M, N, K, \{N_k\}_{k=1}^K$, $\{\hat{\psi}_k\}_{k=1}^K$, $(\mathbf{\Lambda}, \mathbf{U}), d$

- Select the $MN - \sum_k N_k$ eigen vectors of $\mathbf{U}$ corresponding to the least $MN - \sum_k N_k$ eigenvalues to form a basis for noise sub-space, $\mathbf{U}_N$

for $k = 1$ **to** K **do**

for $\phi = -\frac{\pi}{2}$ **to** $\frac{\pi}{2}$ **in steps of** d **do**

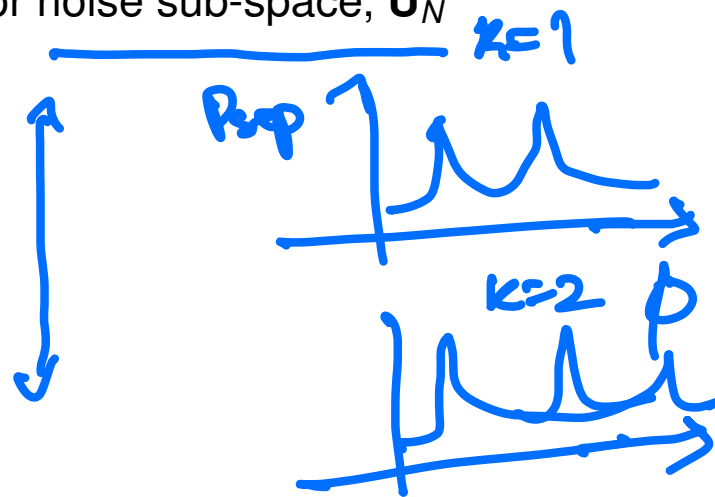
$$\mathbf{P}_{\text{spec}}(k, \phi) = \frac{1}{\| \{ [\mathbf{a}_N(\phi) \otimes \mathbf{I}_M] \mathbf{a}_M(\hat{\psi}_k) \}^H \mathbf{U}_N \|^2_2}$$

end

$$\{\hat{\phi}_{i,k}\}_{i=1}^{N_k} = \underset{\phi}{\operatorname{argmax}}^{(N_k)} \mathbf{P}_{\text{spec}}(k, \phi)$$

end

Output: $\{\hat{\phi}_{i,k}\}_{i=1, k=1}^{N_k, K}$



Algorithm 2: DoD estimation at the UE using MUSIC

³ Can be obtained using channel sounding techniques (or) scree plot methods

Remarks

1 On the number of antennas at BS and UEs:

- To solve subspace methods, we need $MN > \sum_{k=1}^K N_k + 1$ - Not difficult to meet in mmWave mMIMO syms.

2 On the number of pilot transmissions:

- The total # pilot transmissions: NT_2
- To decompose into a full-rank source covariance matrix, $T_2 > \sum_{k=1}^K N_k$
- Comparison with conventional least squares: $NT_2 \ll MN \sum_k L_k$ - **low pilot overhead and not scale with the no. of IRS elements**

$$N \sum_{k=1}^K N_k$$

Result-1

$M = 256$, $N = 2$, $N_k = 1$ for all k , K up to 4, and $L_1 = L_2 = L_3 = L_4 = 512$

$$z = \Delta_{\text{eff}} y \quad \text{Mux } \odot$$

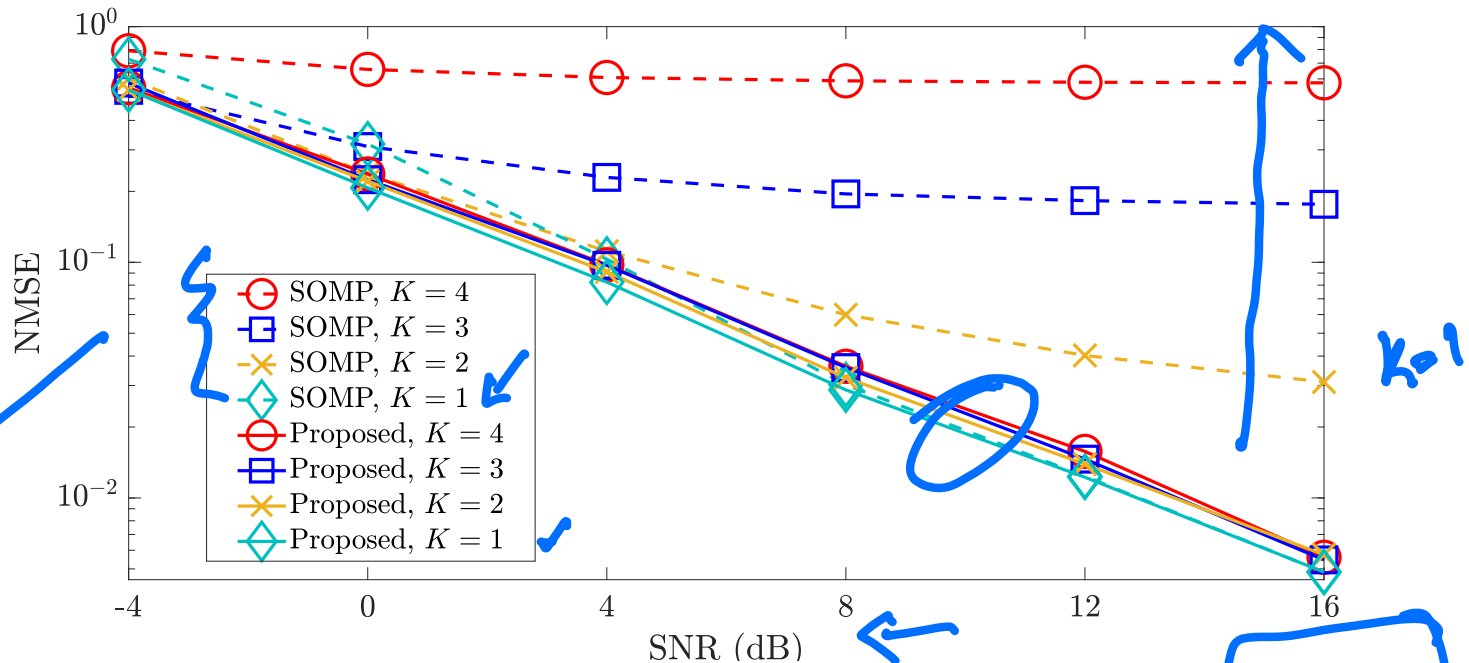


Figure: NMSE vs SNR for $K = 1, 2, 3, 4$

Observation: Robust against the no. of IRSs over conventional approaches!⁴

⁴ SOMP: simultaneous orthogonal matching pursuit; solution for joint sparse recovery problem

Result-2

Sys parameters: same as result-1 except for $K = 2$.

$$N_1 = N_2 = 1$$

$$\sum N_k = 2$$

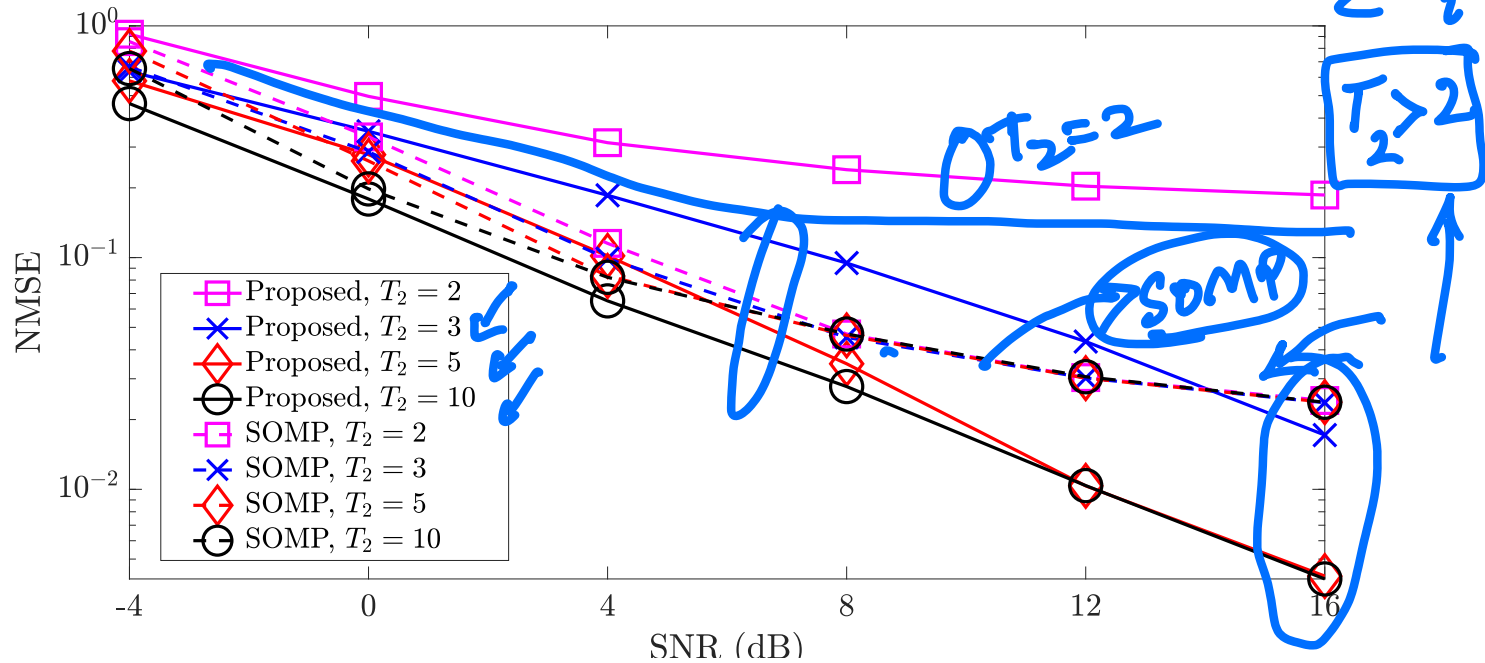
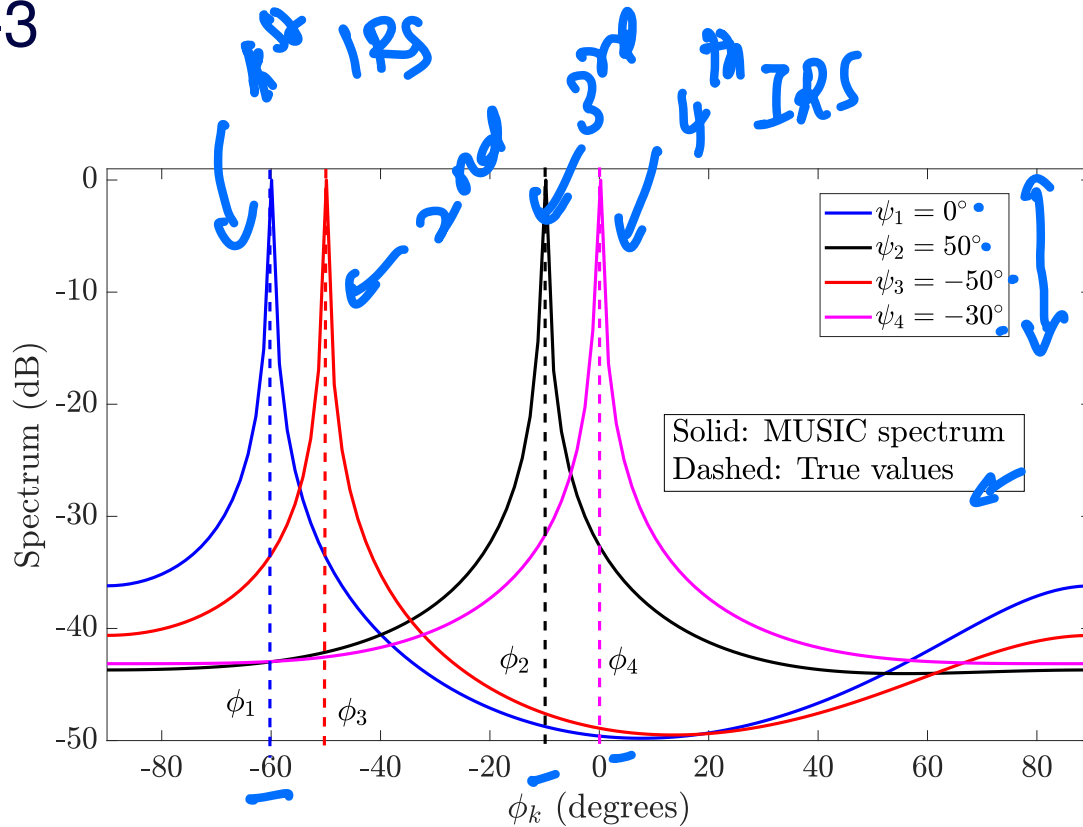


Figure: NMSE vs SNR for different values of T_2

Observation: Performance changes with # pilots and $T_2 > \sum_{k=1}^K N_k$ needs to be satisfied for reasonable performance

Result-3

Figure: MUSIC spectrum at 20 dB SNR for $T_2 = 10$

Observation: DoA-DoD pairing by joint ESPRIT-MUSIC occurs w/o ambiguity

Thank You